BOOTSTRAP AND AUTOCONFIGURATION (DHCP)

Reti di Calcolatori (Modulo 2)

Introduction

- Client-server paradigm used for bootstrapping
- Before a computer can send or receive a datagram, it needs to know
  - **IP address**
  - Address of the router
  - Subnet mask
  - Address of a name server
- Alternative to RARP to determine IP address
  - E.g., used for diskless computers (memory image, file server, IP address of the router)

Recalling RARP

- RARP used to determine an IP address at startup
  - A TCP/IP protocol
  - RARP message encapsulated in the data portion of a network frame (e.g., ethernet frame, type 8035_{16})

Recalling RARP

- Communication flow
  - Sender broadcast a RARP request
  - Sender hardware address is added as the target address
  - Only the computer providing the RARP service (RARP server) process the request
  - A reply is produced (IP address added) and returned to the sender

Recalling RARP: Drawbacks

- Require direct access to the network hardware
  - No possibility of building a server as an application program
- Data exchange is limited to the IP address only
  - Additional information could be sent at no cost
- Cannot be used in networks with dynamic hardware addresses

System Startup

- To keep protocol software general
  - IP stack designed with many parameters
  - Values filled in when system starts

- Two possible sources of information
  - Local storage device (e.g., disk)
  - Server on the network

Bootstrapping

- BOOTstrap Protocol (BOOTP)
  - Early alternative to RARP
  - Provides more than just an IP address
  - Obtains configuration parameters from a server (for diskless computers)
  - Used UDP
- Dynamic Host Configuration Protocol (DHCP)
  - Replaces and extends BOOTP
  - Provides dynamic address assignment

Apparent Contradiction

- BOOTP used to obtain parameters for an IP stack
- BOOTP uses IP and UDP
- Stack must be initialized before being initialized

Problem

- How can a computer send a BOOTP request in an IP datagram before the computer learn its IP address??????

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- How can a computer send a BOOTP request in an IP datagram before the computer learn its IP address??????
- Special-case IP address → Broadcast

Solution

- Use the special IP address with all 1s (255.255.255.255)
- IP software accepts and broadcasts datagrams that specify the limited broadcast address
  - Also if the software has not discovered its IP address yet

Solving the Apparent Contradiction

- BOOTP runs as application
- Only needs basic facilities

In particular:

An application program can use the limited broadcast IP address to force IP to broadcast a datagram on the local network before IP has discovered the IP address of the local network or the machine's IP address.

Solving the Apparent Contradiction

- User A, BOOTP server B
- A needs an IP address and sends a BOOTP request in broadcast
  - Source IP: 0.0.0.0
  - Destination IP: 255.255.255.255 (broadcast)
- What happen for the reply? B cannot use ARP when replying because the client does not know its own IP address

Solving the Apparent Contradiction

- ARP used to map an IP address to the hardware one
  - To do that it sends a broadcast message (with the IP address) and wait the response of the target
  - Not possible, A does not have the IP address

- Alternatives
  - Update server cache with IP_A<->HW_A
  - Response broadcast

BOOTP Retransmission

- Client handles retransmission
  - Message can be delayed, lost, duplicated…
  - IP does not provide checksum to check for corrupted bit
- BOOTP requires UDP with checksum
- BOOTP uses requests/replies with the *do not fragment bit* set

BOOTP Retransmission

- BOOTP uses the technique of *timeout and retransmission*
  - A timer is set
  - If no response is received after the timer is over the message is retransmitted
  - An initial timeout is selected at random (0-4 sec)
  - Timeout for successive retransmissions doubled (to avoid excessive traffic)
  - After 60 sec only randomization

BOOTP Message Format

- Fixed length fields
- Both request and response have the same format
- Similar to the one used for DHCP

Two-Step Bootstrap

- BOOTP provides information, not data
- Client receives
  - Fully qualified name of file that contains boot image (BOOT FILE NAME)
  - Address of server
- Clean separation between configuration and storage
- Client must use another means to obtain the image to run (typically TFTP)

BOOTP: Problems - 1

- Designed for a static environment
  - Each host has a permanent network connection
  - Each host has an entry in the BOOTP configuration file
- Not suitable for dynamic environment
  - Portable and wireless devices with no fixed positions
  - Configuration information changes quickly

BOOTP: Problems - 2

- BOOTP provides static mapping host<->host parameters
- Mapping stored in BOOTP server configuration file
- No dynamic configuration
  - A manager assigns a host identifier to its IP address

BOOTP: Problems - 3

- Not suitable when computers change place often and are more than the number of available IP addresses
- Suppose that a LAN have
  - Class C addressed (254 hosts max)
  - 10 groups of 30 users (e.g., 300 users max)
  - Max hosts < max users
  - A manager cannot assign a unique address to all users
  - He needs to change the configuration file at each access

Dynamic Host Configuration Protocol (DHCP)

- Dynamic Host Configuration protocol designed by IETF
  - Dynamic address assignment
- Two main differences with BOOTP
  - Acquire all necessary information in a single message
  - Allow a computer to obtain an IP address quickly and dynamically

Dynamic Address Assignment

- Managers configure a DHCP server with a set of IP addresses
- Needed by ISPs (alternative to PPP)
  - Client obtains an IP address and uses it temporarily
  - When client finishes, address is available for another client
- Used by almost all corporate networks

DHCP Address Assignment

- Backward compatible with BOOTP
- Can assign addresses in three ways
  - Manual (manager specifies binding as in BOOTP)
  - Automatic (address assigned by server, and machine retains same address)
  - Dynamic (address assigned by server, but machine may obtain new address for successive request)
- Manager chooses type of assignment for each address, each network, each host

DHCP Address Assignment

- Manual (manager specifies binding as in BOOTP)
  - Static allocation, the network administrator chooses the IP address to assign to the client and the DHCP server sends it to the client
  - A static DHCP allocation is permanent; it is done by configuring a DHCP server and choosing a Reserved Address to correspond to the MAC Address of the client device
  - The DHCP assignment remains in place even if the client logs off, reboots, has a power outage, etc.

DHCP Address Assignment

- Automatic (address assigned by server, and machine retains same address)
  - The DHCP server assigns a permanent IP address to a client from its IP Pools
  - Lease specified as Unlimited means the allocation is permanent
- Dynamic (address assigned by server, but machine may obtain new address for successive request)
  - The DHCP server assigns a reusable IP address from IP Pools of addresses to a client for a maximum period of time (lease)

DHCP Support For Autoconfiguration

- *Because it allows a host to obtain all the parameters needed for communication without manual intervention, DHCP permits autoconfiguration. Autoconfiguration is, of course, subject to administrative constraints.*

Dynamic Address Assignment

- It is the most significant paradigm shift
- The server does not need to know each user a-priori
- DHCP allows systems that autoconfigure
  - The user connects to the network
  - The user receives the IP address from DHCP
  - The user configures his TCP/IP stack
- Autoconfiguration subject to administrative rules

Dynamic Address Assignment

- Dynamic Address Assignment is temporary
  - Client is granted a *lease* on an address
  - Server specifies length of lease
  - At the end of lease, client must renew lease or stop using address

- Actions controlled by finite state machine

Dynamic Address Assignment

- Length of the lease depends on the application context
  - It can be short (e.g., 1 hour in a university laboratory)
  - It can be long (e.g., 1 week in a corporate network)
  - It can be infinite
- The client can require a lease period, the server can accept or provide a different lease period

Δ BALANCE Performance / Space use

Multiple Addresses

- A computer may connect to more that one network (multiple interfaces)
- Each interface must be handled separately
- A single DHCP message for a single interface
- Different interfaces at different points of the protocol

Server Contact

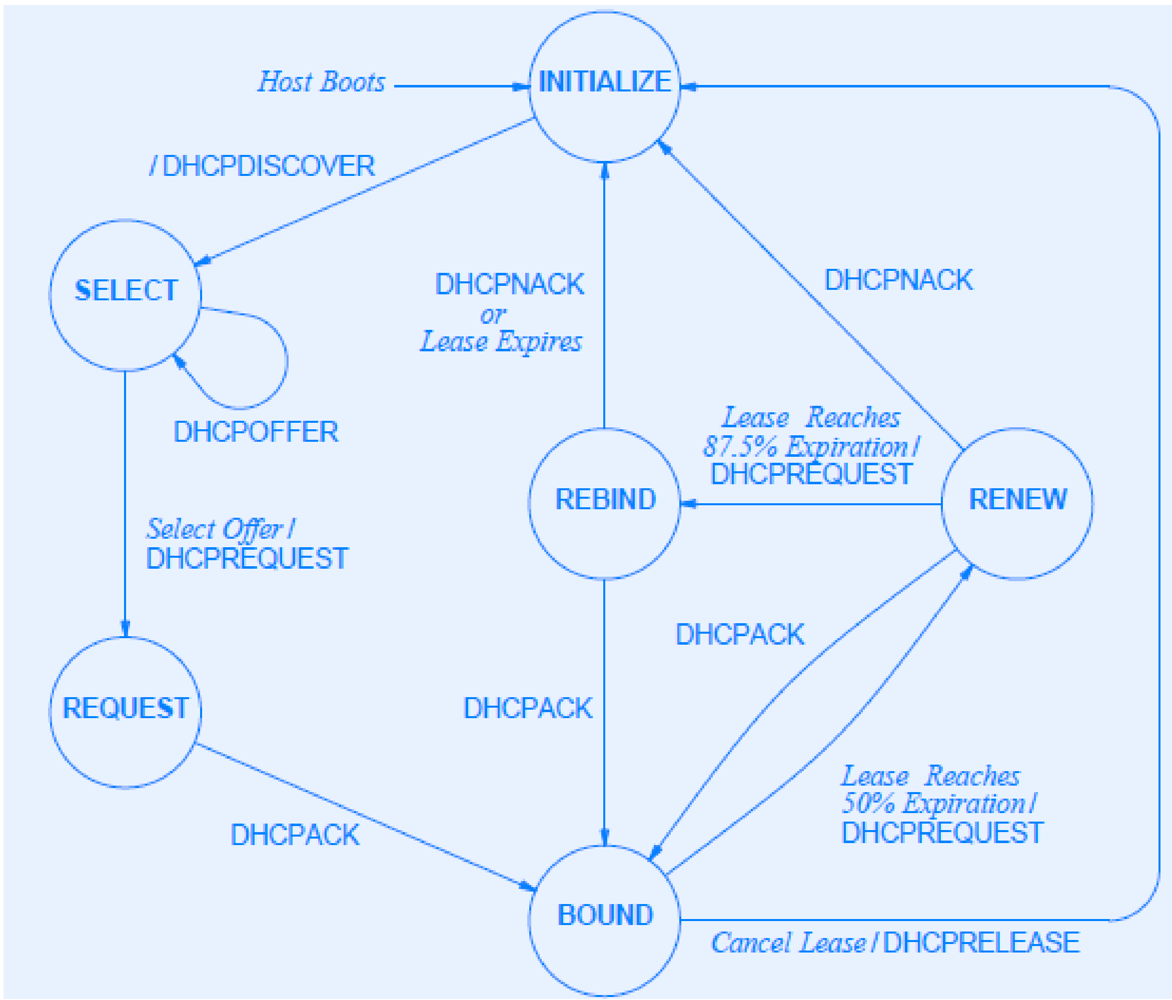
- *To use DHCP, a host becomes a client by broadcasting a message to all servers on the local network. The host then collects offers from servers, selects one of the offers, and verifies acceptance with the server*
- *Afterwards the host may require renewal of the offer*

Protocol Flow

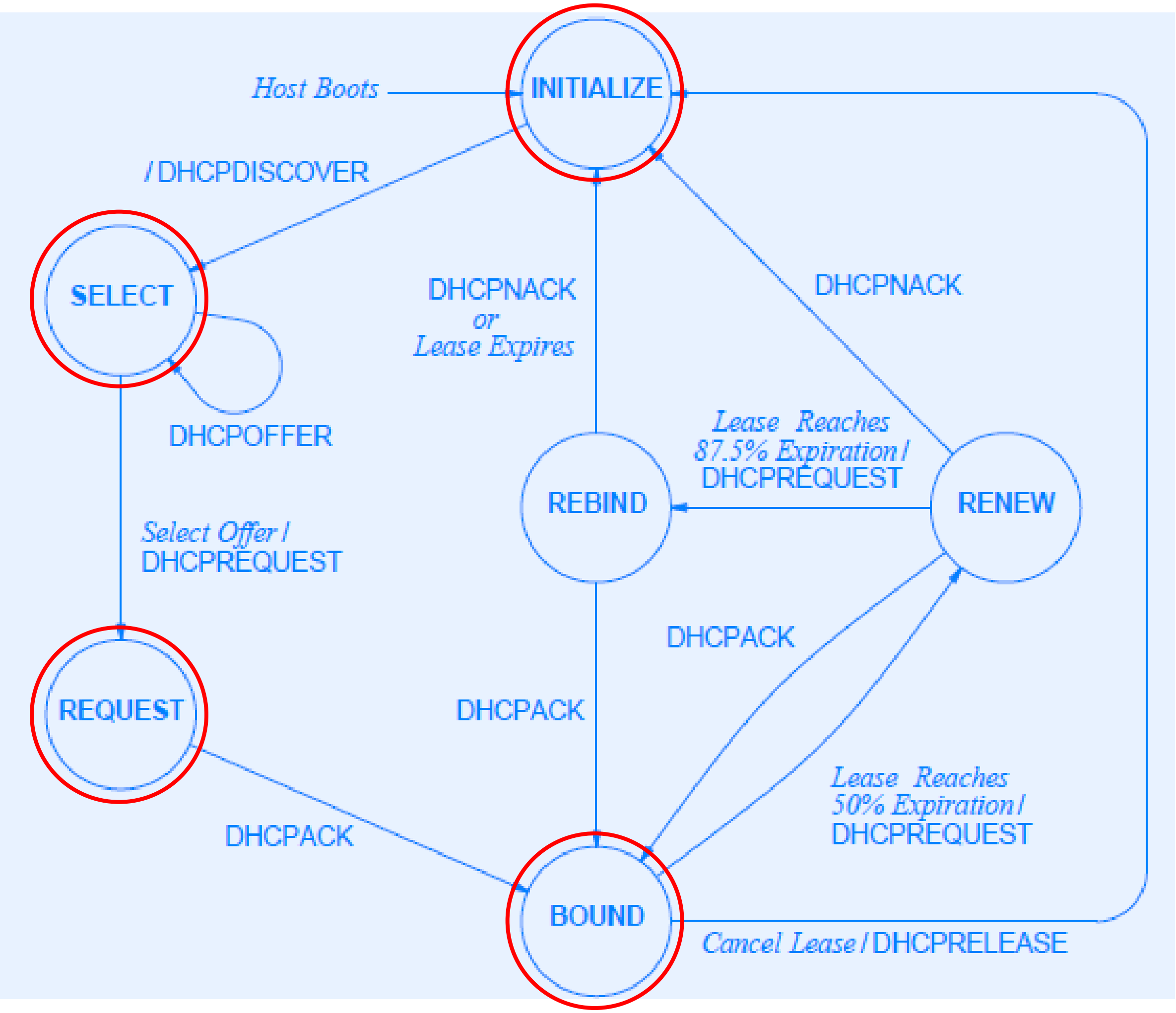
- 6 states, 11 possible transitions

- Three main interactions
  - Address acquisition
  - Lease renewal state
  - Early release termination

DHCP Finite State Machine



DHCP Finite State Machine: Address Acquisition



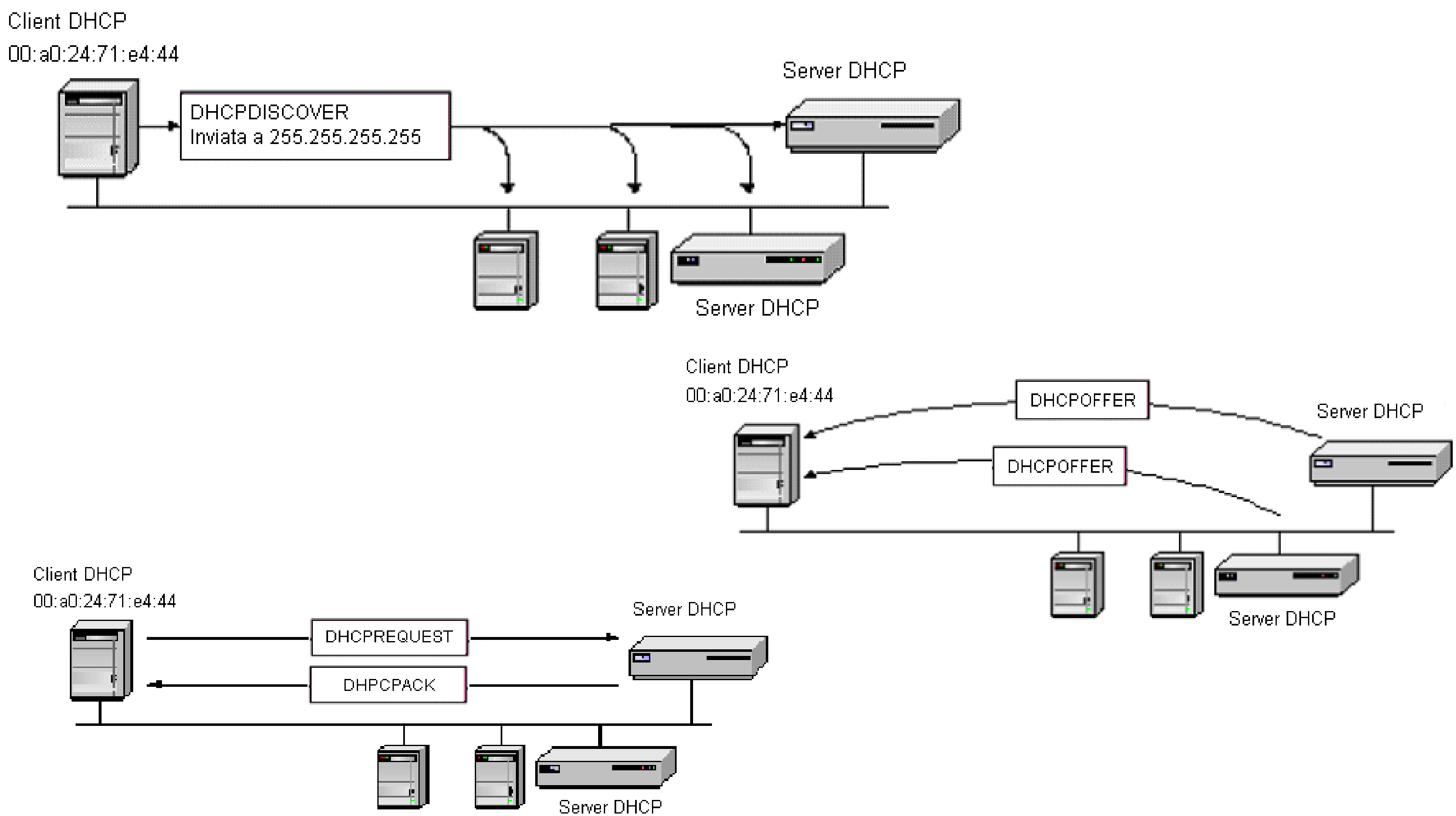
Protocol Flow: Address acquisition - 1

- The client boots and enters the INITIALIZE state
- The client contacts the DHCP servers
  - Send a DHCPDISCOVER and moves to SELECT state
  - Extension of BOOTP, send a broadcast UDP message to BOOTP port 67
- The servers receive the message and reply with a DHCPOFFER message

Protocol Flow: Address acquisition - 2

- The client collects the DHCPOFFER messages
  - Each offer contains configuration information (including IP address)
- The client chooses one DHCPOFFER and negotiates for a lease
  - Send a DHCPREQUEST
  - Enter REQUEST state
- The server acks the request and the lease starts (DHCPACK message)
  - The client enters BOUND state

Protocol Flow: Address acquisition - 3

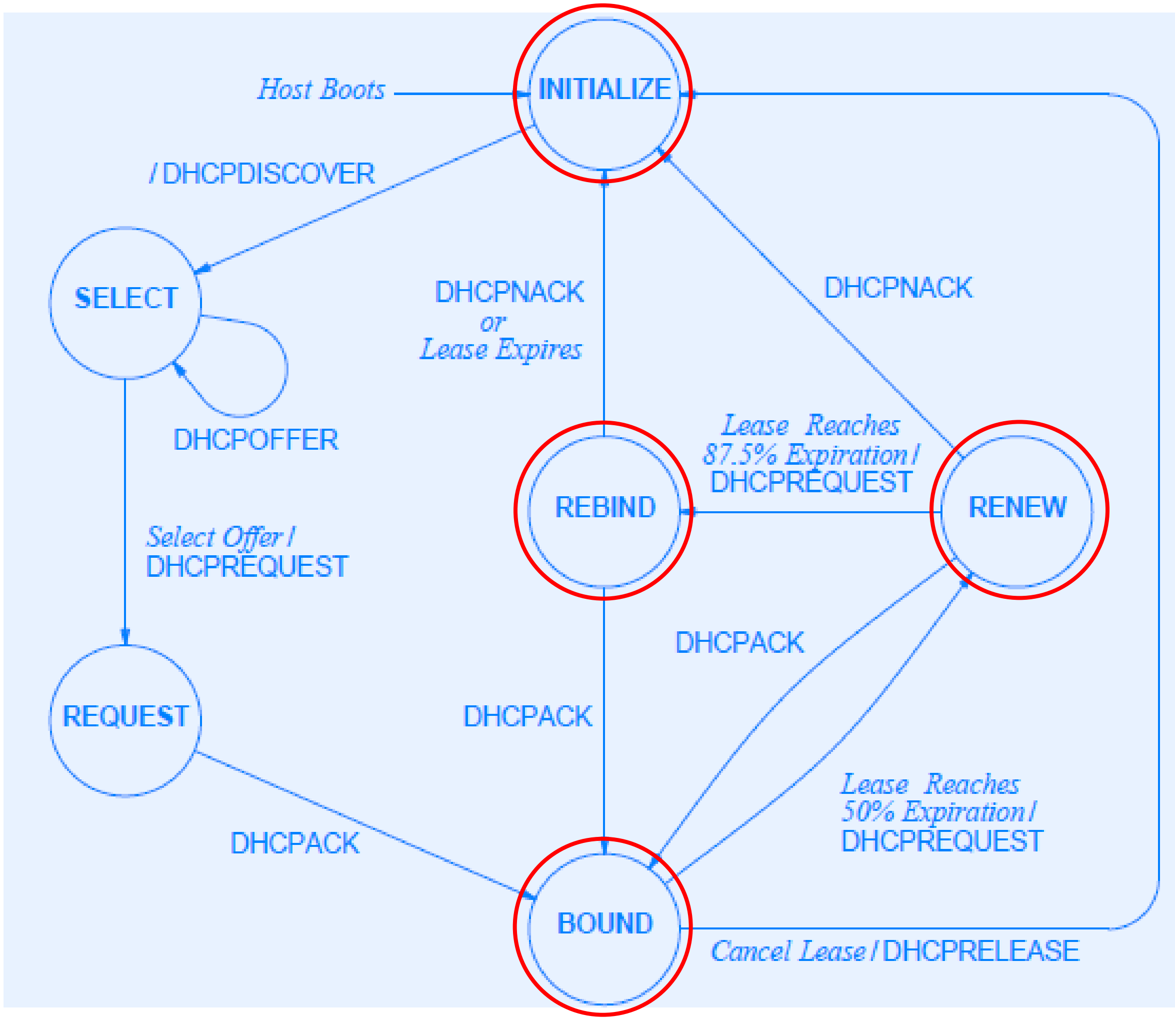


Lease Timers

- When clients are in the BOUND state, 3 timers are set by the server

- Default values
  - Renewal = 0.5 * total_lease_time → Client chiede rinnovo al server
    - → Sul server che ha inviato e' offerta
  - Rebinding = 0.875 * total_lease_time → Contatta server alternativo
    - → Se a Renewal non riceve risposta
    - → In broadcast
      - → client non conosce il server alternativo
  - Expiration = 1 * total_lease_time
    - → Rilascio

DHCP Finite State Machine: Lease Renewal State



Protocol Flow: Lease Renewal State - 1

- When renewal timer expires
  - The client tries to renew the lease
  - It sends a DHCPREQUEST with the IP address and enters the RENEW state
  - If the renew is approved, the server sends a DHCPACK and the client enters the BOUND state
  - If not the server sends a DHCPNACK and client enters the INITIALIZE state

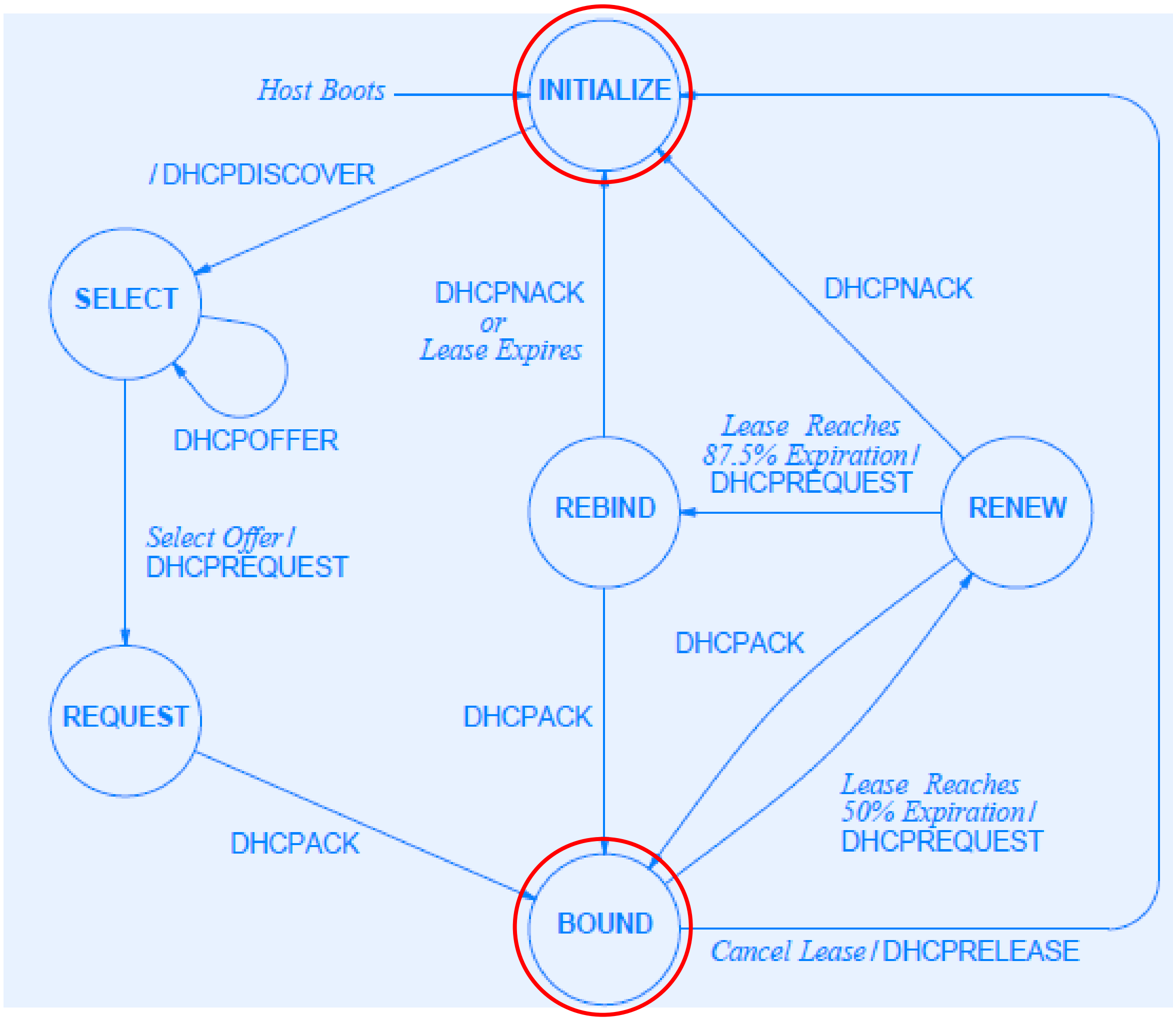
Protocol Flow: Lease Renewal State - 2

- If the server is down or unreachable, rebinding timer is used
- The client moves from RENEW to REBIND state
- The client broadcasts a DHCPREQUEST to each server on the Net
- If the renew is approved by one of the servers, the client enters the BOUND state
- If not, the client stops using the IP address and enters the INITIALIZE state

Protocol Flow: Lease Renewal State - 3

- If no server replies to the client, the expiration timer expires
- The client release the IP address and moves to the INITIALIZE state

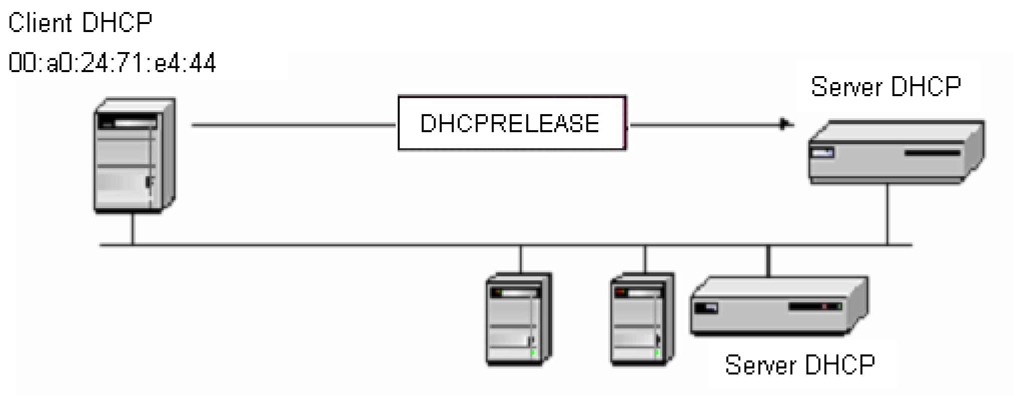
DHCP Finite State Machine: Early Release Termination



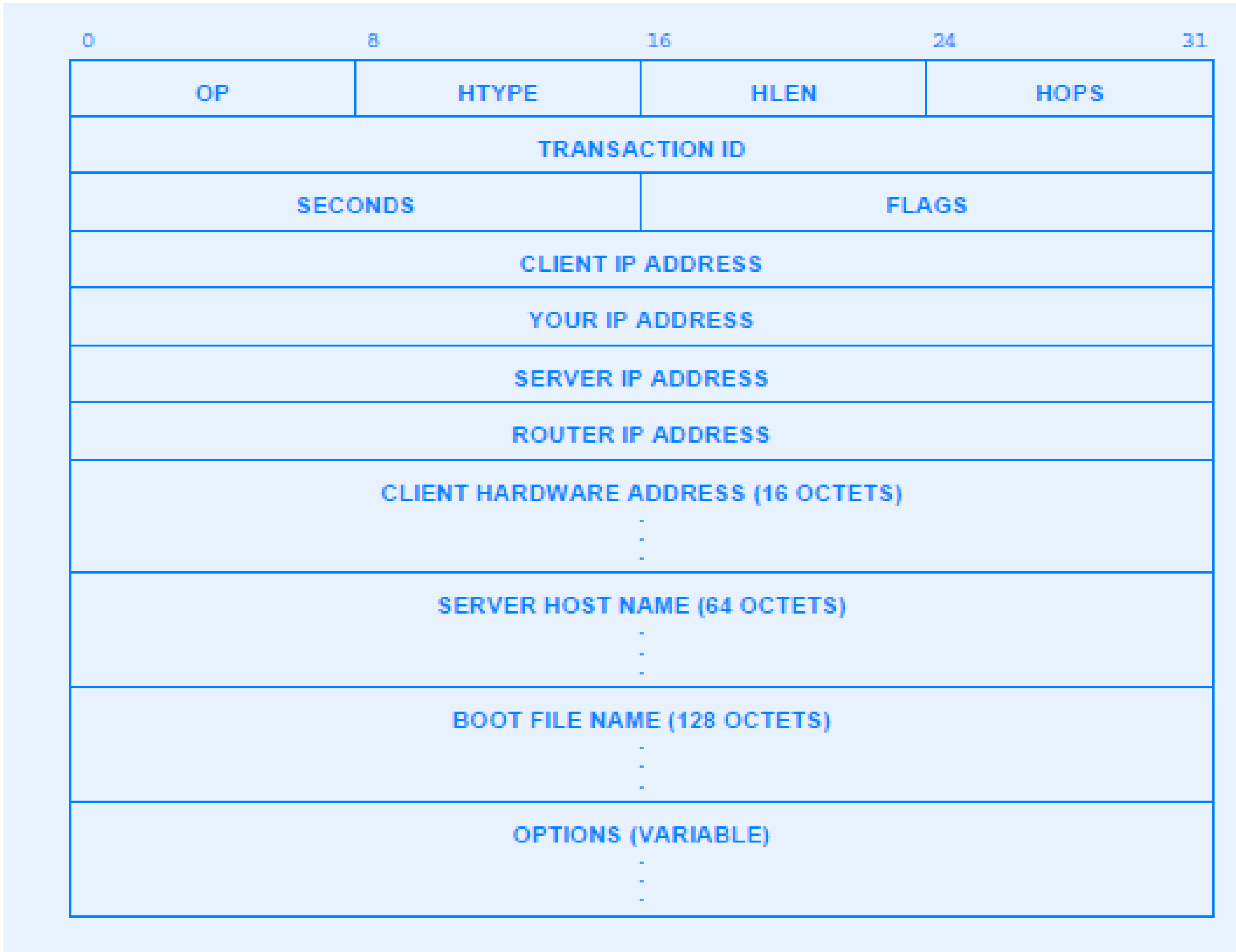
Protocol Flow: Early Release Termination

- Clients can store the IP address in the memory to request the same at each boot
- What happen when a client disconnect?
  - Minimum lease period (e.g., 1 hour)
  - Terminate a lease explicitly by sending DHCPRELEASE message and moving to INITIALIZE state
  - Help when there are many users and few addresses

Protocol Flow: Early Release Termination



DHCP Message Format



DHCP Message Format

- Similar to BOOTP message (for backward compatibility)
  - OP: message request (1) or reply (2)
  - HTYPE: network hw type (type 1 for ethernet)
  - HLEN: address length (6 octect for ethernet)
  - HOPS: usually =0
  - TRANSACTION ID*:* A 32-bit identification field generated by the client, to allow it to match up the request with replies received from DHCP servers

DHCP Message Format

- Similar to BOOTP message (for backward compatibility)
  - SECONDS: the number of seconds elapsed since a client began an attempt to acquire or renew a lease
  - * ADDRESS: all 0s if unknown
  - SERVER HOST NAME: name of the specific server
  - BOOT FILE NAME: to identify the boot file

DHCP Message Format

- Differences with BOOTP message
  - UNUSED -> FLAGS

0	15
B	MUST BE ZERO

  - 16-bit FLAGS field: only the high order bit as meaning
  - Leftmost bit is 0 for hardware unicast, is 1 for hardware broadcast

DHCP Message Format

- Differences with BOOTP message
  - VENDOR SPECIFIC AREA -> OPTION
  - Encode information such as lease duration, subnet mask, server and host name, domain name

- *"Option Overload"* allows OPTION field to use SERVER HOST NAME and BOOT FILE NAME if empty

Message Type Field

0	8	16	23
Code (53)	Length (1)	Type (1-8)	

TYPE FIELD	Corresponding DHCP Message Type
1	DHCPDISCOVER
2	DHCPOFFER
3	DHCPREQUEST
4	DHCPDECLINE
5	DHCPACK
6	DHCPNACK
7	DHCPRELEASE
8	DHCPINFORM

Relay Agents

- *Relay agent* permits a computer to contact servers on a nonlocal networks
  - Receives a request from the client (broadcast)
  - Forwards the request to the server
  - Sends the response from the server to the client

- Multiple interfaces mean multiple requests
  - The server distinguishes the interfaces using the hardware address

Relay Agents

- How the DHCP can know the address to assign to the client being out of client's network?
- The relay agent stores its own IP address in the ROUTER IP ADDRESS (aka GIADDR) → Gateway ip addr
- DHCP server determines the subnet of the client and allocates an IP address on that subnet

Summary

- Two protocols available for bootstrapping
  - BOOTP (static binding of IP address to computer)
  - DHCP (extension of BOOTP that adds dynamic binding of IP addresses)
- DHCP
  - Server grants lease for an address
  - Lease specifies length of time
  - Host must renew lease or stop using address when lease expires
  - Actions controlled by finite state machine

Esercizio 1

- Si supponga che un amministratore di rete debba gestire una rete locale con indirizzi di classe C (massimo 254 host). La rete locale deve distribuire gli indirizzi a 50 macchine fisse e sempre collegate alla rete
- Discutere le diverse soluzioni applicabili

Esercizio 2

- Si supponga che un amministratore di rete debba gestire una rete locale con indirizzi di classe C (massimo 254 host). La rete locale deve distribuire gli indirizzi a 5 gruppi da 100 utenti che accedono alle macchine in momenti diversi della giornata.
- L'amministratore decide di utilizzare il protocollo BOOTP. Come vengono assegnati gli indirizzi? Quale problematica introduce una soluzione basata su BOOTP?
- Come si comporterebbe il sistema se si adottasse una soluzione basata su DHCP?